Journal of Biomechanics 198 (2026) 113192

Contents lists available at ScienceDirect

Journal of Biomechanics

journal homepage: www.elsevier.com/locate/jbiomech

Short communication

Evaluation of automatic identification of events during the lunge exercise using kinematic data

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1. Introduction

Optical motion capture systems are the gold standard for tracking human movement and are increasingly used in rehabilitation to identify injury risks (Van der Kruk and Reijne, 2018). Proper biomechanical analysis requires segmenting movements into phases based on specific events. In gait studies, force plate data are the benchmark, with initial contact (IC) and terminal contact (TC) identified through the onset and cessation of force (Zahradka et al., 2020). However, integrating force plates with motion capture systems is costly and complex, making their use in clinical settings challenging.

To address this, researchers have focused on using kinematic data to identify key movement events. Zahradka et al. reviewed methods for identifying gait events using kinematics, finding that velocity differences between body segments correlated well with specific gait events. Our lab's recent study supported this, showing strong agreement between kinematic and kinetic data for variables like gait speed, stride length, and cycle time in healthy adults (Ripic Set al., 2022). These findings suggest that kinematic data can reliably evaluate movement patterns without the need for force plates.

The lunge, a common movement in rehabilitation, is typically analyzed using force plates to detect the onset and cessation of force as the movement begins and ends (Goulette et al., 2021; Kritz et al., 2009; Marchetti et al., 2018; Nadzalan et al., 2021) However, this method has limitations. It excludes data from the swing phase of the movement, and backward lunges, where the subject is not consistently on the force plate, are not fully captured. Therefore, the purpose of this study was to determine if kinematic data could be an effective way to identify lunge events.

2. Methods

2.1. Subjects

A total of 34 healthy university students (men = 21, women = 13) were recruited. Subjects ranged from 18-30 years of age (mean: 21.62, SD: 3.14 years – See Table 1 of the Appendix for full subject characteristics), were recreationally active, and were free of any musculoskeletal injuries for the previous 6 months prior to testing. This study was approved by the University's Institutional Review Board and the testing protocol, benefits, and risks of the study were explained to the participants prior to them reading and signing a university-approved informed consent.

2.2. Testing procedure

Subjects were instructed to arrive in tight-fitting or compression clothing. If subjects did not have proper clothing, they were provided with a set of compression clothing from the laboratory. Following the informed consent, subjects' height and body mass was measured. Next, subjects' dominant leg length was measured according to established best practices (Sabharwal and Kumar, 2008). A total of 54 reflective markers were placed on the specific anatomical landmarks according to a modified plug-in gait model by Vicon (Vicon LLS, Oxford, UK) with an additional marker on each lower body segment to ensure a minimum of three markers were visible by the camera system at all times (Fig. 1)

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https://doi.org/10.1016/j.jbiomech.2026.113192
Accepted 28 January 2026
Available online 30 January 2026
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(Ripic et al., 2022).

A 10-camera motion capture system (Qualysis AB, Gothenburg, Sweden) and two adjacent force plates (Type 9286AA; Kistler Instrument AG, Winterthur, Switzerland) were used to track the markers and ground reaction forces during the execution of the lunges. Calibration of the system was in accordance with the directions in the user manual. Reconstructed static trials were visually inspected to ensure all markers were present. Frame capture rate was set at 100 frames per second (fps) and force sampling rate was set to 1000 Hz.

Subjects were then asked to complete the trials by performing five lunges forward to a marked distance equaling 75% of the subject's dominant leg length, in accordance with previous research (Goulette et al., 2021). Subjects were asked to keep the heel of the dominant leg on the ground at all times when lunging to prevent excessive patellofemoral stress (Chandler et al., 2002; Zellmer et al., 2019).

2.3. Data analysis

2.3.1. Marker-based 3DMA

Marker-based files were transferred from the collection system to an offline processing station. Trials were reviewed to ensure that all markers and identified body segments were present during each trial and the captured segments were cropped to include only the lunge movement. Missing markers were gap-filled using a relational function based on three other markers present on the same body segment. In cases where three markers were not present, a single marker on the same segment was used to gap-fill; and in cases where less than 10 frames were missing, gaps were filled using cubic spline fitting. Processed files containing marker and force plate data were then imported into Visual 3D (C-Motion Inc., Germantown, MD) for skeletal modeling and lunge event determination. Force plate data was automatically down sampled to match marker sampling, which is a standard step in Visual 3D for inverse dynamics analysis.

2.3.2. Lunge event determination

Based on the previous research by (Zahradka et al., 2020), we elected to identify the lunge events as follows: the start of the lunge occurred when the velocity of the heel of the nondominant leg (i.e. the leg that was always the trail, or back leg) relative to the pelvis exceeded −0.2 m. s^{-1} in the anteroposterior direction, the bottom of the lunge was identified as the minimum value for the position of the pelvis in the vertical direction, and the end of the lunge was defined as the threshold crossing of the anteroposterior heel velocity relative to the pelvis at 0.2 m.s^{-1}. This was done in accordance with previous work done in this laboratory, which used similar methods to identify gait events using only kinematic data of the pelvis and foot segments (Ripic et al., 2022).

Because of the low velocity of the movement, pelvis center of mass and heel landmarks were lowpass filtered at 1 Hz to eliminate any spikes or artifacts. Thresholds of vertical ground reaction force were set at 20 N to determine the beginning and end of the lunge, which is similar to the threshold that is typically used during gait event determination (Ripic et al., 2022; Zahradka et al., 2020; Zeni et al., 2008). Due to the risks associated with low frame rates and temporal smearing, a sensitivity analysis was conducted on data collected and is provided in Table 2 of the Appendix.

2.4. Statistical analysis

Kinematic and kinetic data were averaged across five trials for each subject, and root mean square errors and standard deviations for lunge events were calculated. To assess agreement between systems, Type C two-way mixed effects intraclass correlation coefficients (ICCs) were used, where values under 0.5 indicated poor agreement, 0.5–0.75 indicated moderate agreement, 0.75–0.9 indicated good agreement, and values over 0.9 indicated excellent agreement (Koo and Li, 2016). Additionally, Pearson correlation coefficients evaluated the similarity between methods, with values ranging from 0.0 to 0.3 considered negligible, 0.3–0.5 low, 0.5–0.7 moderate, 0.7–0.9 high, and 0.9–1.0 very high (Mukaka, 2012). Paired t-tests determined statistical differences between means, while Bland-Altman analysis assessed the limits of agreement between systems. Due to limited research on the validity of kinematic events during lunges, results from the Bland-Altman analysis were compared to previous studies that examined kinematically-defined events (Ripic et al., 2022; Zahradka et al., 2020; Zeni et al., 2008). Statistical analyses were conducted using SPSS version 29.0.1.0.

3. Results

A total of 67 trials containing matched kinematic and kinetic data for the start of the lunge and 95 trials containing matched kinematic and kinetic data for the end of the lunge were collected and included in the analysis. Results for the comparisons of lunge event determinations between kinematic and force plate are shown in Table 1. ICCs indicated excellent agreement between event determination methods. Paired t-tests showed significant differences between the two methods for the start point ($p < 0.001$) and end point ($p = 0.036$) of the lunge.

Results of the Bland Altman analysis can be found in Table 2. Biases between systems for the start (−133 ± 214 ms) and end (184 ± 844 ms) of the exercise were small, each being less than one percent of the maximum scores. The Bland-Altman plots for the start and end points of the lunge with a regression plot showing the relationships between methods are presented in Figs. 1 and 2 of the Appendix, respectively.

4. Discussion

This study aimed to explore a novel method for detecting key events during the lunge exercise using kinematic data and comparing it to the gold standard kinetic method, which relies on force plates. While significant differences were observed between the mean scores of both methods, correlation and intraclass analyses demonstrated a strong relationship between the two techniques, with Bland-Altman analyses showing excellent agreement, especially at the lunge's endpoint.

The high correlations between the two methods are similar to findings by Ripic et al. (2022), who validated kinematically determined gait events against a markerless motion capture system integrated with force plates. They reported high intraclass correlations (ICC = 0.909–0.997)

Table 1
T-test and correlation results for comparisons between force plate and kinematic data during the lunge exercise.

	Kinematic Mean ± SD (ms)	Force Plate Mean ± SD (ms)	MDiff ± SE (ms)	95%CI (ms)	p[†]	d	r	ICC (95%CI)	ICC (95%CI)
Start (n = 67)	7030 ± 4260	7170 ± 4200*	−130 ± 20	−190, −80,	0<.001	−0.033	0.999	0.998 (0.998, 0.999)	0.998 (0.998, 0.999)
End (n = 95)	7430 ± 4860	7710 ± 4880*	−20 ± 10	−40, 0	0.036	−0.057	1	1.00 (1.00, 1.00)	1.00 (1.00, 1.00)

All values are in ms. SD = standard deviation; SE = standard error; MDiff = mean difference between the two conditions; p = probability derived from the *t*-test, d = Cohen's d; r = Pearson's correlation coefficient; ICC = intraclass correlation coefficient; 95%CI = 95% confidence interval.
*significant difference between conditions.

for most spatial and temporal variables, except for the swing phase (ICC = 0.675) and double-leg stance (ICC = 0.701). In our study, relative heel velocity was used to define the start and end of the lunge, yielding even higher agreement (ICC = 0.998–1.000). The differences in agreement could be due to variations in movement patterns, velocities, or data collection methods. Unlike Ripic's study, which lacked secondary toe and heel markers, our model included these markers, allowing for better foot position estimates.

Zahradkha and colleagues (2020) similarly compared kinetic and kinematic data to identify gait events, finding that foot velocity detected initial contact 0.495 s earlier and terminal contact 0.069 s later than the force plate. In our study, the force plate detected the start of the lunge 0.133 s later and the end 0.018 s sooner than the kinematic method. Differences in movement types, equipment, and analysis methods likely contributed to these variations. Zahradka's study calculated foot velocity based on the center of mass, whereas we measured it using heel markers, which could have impacted the results. Additionally, our event detection threshold of ±0.2 m/s, selected during pilot testing, may have introduced some error. Bland-Altman analysis showed better agreement at the lunge's endpoint than at its start. For the start, the mean difference between methods was less than 2%, with only six out of 67 data points falling outside the limits of agreement. At the endpoint, agreement was stronger, with just two out of 95 data points outside the limits of agreement. These findings suggest that while statistically significant differences exist between the two methods, the variation between our event determination method and the gold standard appears smaller than previous studies, which have been utilized across varying populations (Aftab and Shad, 2022; Ho et al., 2022). The practical significance of these results show that kinematic data can reliably detect lunge events, though agreement is better for the endpoint.

This study adds to existing literature and supports the theory that kinematically defined events can be another useful method for the assessment of movement. Markerless motion capture has increased in popularity, in part, due to the elimination of the requirement for marker placement, but also in response to the increased attention placed on biomechanical analysis of athletes. One potential benefit of markerless systems is portability and not being confined to a laboratory space, which may restrict certain movements. The results of this study support the use of these systems and analysis of movements in the field or in laboratory spaces that do not have force plates. By effectively identifying events, further research might examine spatiotemporal data or relative joint velocities that occur between lunge events.

Table 2
Results for the Bland-Altman bias and LOA comparisons between force plate and kinematic data for forward lunge events.

	Bias ± SD	Bias 95% CI	LOA	Lower LOA, 95% CI	Upper LOA, 95% CI
Start n = 67)	−133 ± 214	−185, −81	−552, 286	−420, −620	196, 376
End (n = 95)	18 ± 84	−36, −00	−184, 147	−214, −154	117, 177

All values are in ms. LOA = limits of agreement; CI = confidence interval. Start = start of the lunge exercise; End = end of the lunge exercise.

This study has several limitations. First, there is little previous research on the biomechanics of the lunge, as it is typically used for musculoskeletal strengthening in clinical settings rather than for lower body function analysis. Most prior studies focused on muscle activity or joint loading (Begalle et al., 2012; Boudreau et al., 2009; Distefano et al., 2009; Ekstrom et al., 2007; Escamilla et al., 2022; Keogh, 1999; Kritz et al., 2009). To our knowledge, this is the first attempt to segment lunges based on kinematic data.

Second, the study's use of a lower filtering rate due to the slower movement of lunges compared to walking or running may have influenced the accuracy of event detection. Although excellent agreement was observed, the lower rate may have smoothed over the true start and end points. Future studies should explore the impact of higher filtering rates.

Finally, our results are limited to young, healthy adults, and a specific technique that emphasized the heel of the lead leg remaining on the ground, which may not be generalizable to other populations. Further research should assess different demographics to improve external validity. Additionally, there are several variations and progressions of the traditional lunge, like the walking lunge, where cyclical movement of both legs result in an alternating anchor or stabilizing leg. Subsequent studies may wish to validate this methodology across different lunge patterns.

5. Conclusion

This study validated the use of kinematic data to identify lunge events, showing excellent agreement and minimal bias compared to force plates. Kinematic data provide a reliable alternative for biomechanical analysis of lunges when force plates are unavailable, offering a

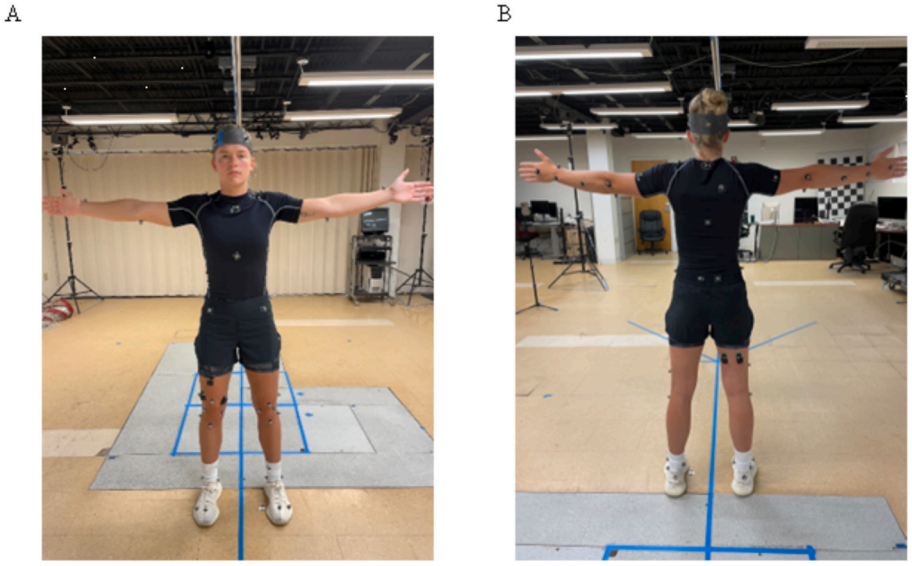


Fig. 1. Reflective marker placement for the (A) front and (B) back.

novel method for movement screening.

CRediT authorship contribution statement

Mitchell Nienhuis: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Joseph Signorile:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **Zachary Ripic:** Writing – original draft, Project administration, Methodology, Formal analysis, Data curation. **Brandon Schoenwether:** Methodology, Formal analysis, Data curation. **Soyeon Ahn:** Project administration, Methodology, Formal analysis. **Kevin Jacobs:** Project administration, Investigation, Conceptualization. **Moataz Eltoukhy:** Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jbiomech.2026.113192.

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